

HW 13

13.1, 13.4, 13.7, 13.11, 13.13

13.4 referencing 13.17 model →

$$D_1 = Q_2 Q_3 \quad D_3 = Q_2 + X$$

$$D_2 = Q_3 \quad Z = X Q_2 + X' Q_2$$

Note: look at D_1, D_2, D_3 as Q_1, Q_2, Q_3

a) Construct a transition table and state graph for a circuit.

Note: this is generally gonna be an 8 bit table

	$Q_3 Q_2 Q_1$	$X=0$	$X=1$	$Z=0$	$Z=1$
S0	0 0 0	0 0 1	0 0 1	0	1
S1	0 0 1	0 1 1	0 1 1	0	0
S2	0 1 0	1 0 0	1 0 1	1	0
S3	0 1 1	1 0 0	0 1 1	1	0
S4	1 0 0	0 0 1	0 0 1	0	0
S5	1 0 1	0 1 1	1 1 1	0	1
S6	1 1 0	1 0 0	0 0 1	1	0
S7	1 1 1	0 1 1	1 1 1	1	0

← $Z=1$ is the inverse of $Z=0$

Notice how the $Z=0$ is following Q_2 pattern will repeat after the fourth seq.

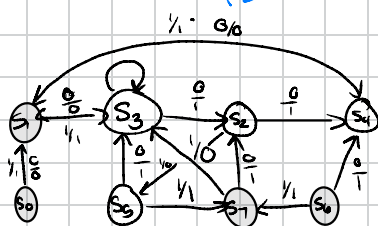
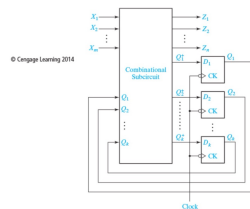


Figure 13-17 General Model for Mealy Circuit Using Clocked D Flip-Flops



Details

13.4c)

01(0)101 ← diagram
01101 ← graph

d) it should change input on falling edge of the clock.

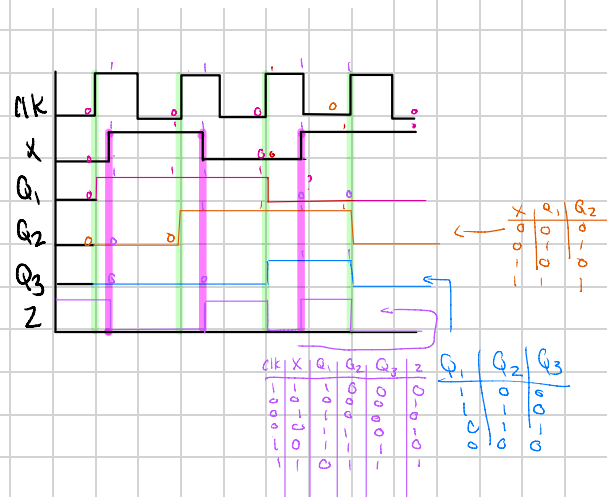
13.13

13.13 A sequential circuit has one input X , one output Z , and three flip-flops Q_1 , Q_2 , and Q_3 . The transition and output tables for the circuit follow:

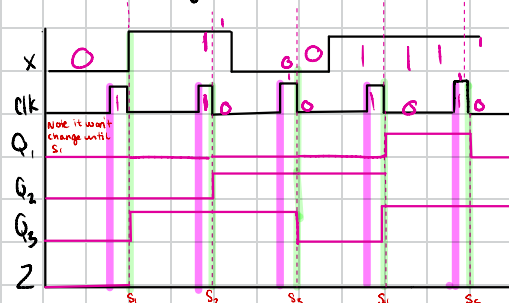
	Next State			Output (Z)	
	$X=0$	$X=1$	$X=0$	$X=1$	
S0	000	100	101	1	0
S1	001	100	101	0	1
S2	010	000	000	1	0
S3	011	000	000	0	1
S4	100	111	110	1	0
S5	101	110	110	0	1
S6	110	011	010	1	0
S7	111	011	011	0	1

a. Construct a timing chart for the input sequence $X = 0101$ and initial state $Q_1 Q_2 Q_3 = 000$. Identify any false outputs. (Assume that the flip-flops are rising-edge triggered and that the input changes midway between the rising and falling edges of the clock.)

b. List the output values produced by the input sequence.



b) Time diagram



$X=0, \text{clk}=0, Q=0$
 $X=1, \text{clk}=0, Q=0$

Q_1

X	clk	Q_1
0	1	0
1	0	0
0	1	0
1	0	0

Q_2

X	Q_1	Q_2
0	0	0
1	0	0
0	0	0
1	0	0

Q_3

X	Q_1	Q_2	Q_3
0	0	0	0
1	0	0	0
0	0	0	0
1	0	0	0

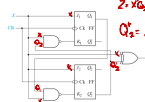
$Z=$

13.7

13.7 a. Construct a transition table and state graph for the circuit shown.

b. Construct a timing chart for the circuit for an input sequence $X = 1011$. Assume that initially $Q_1 = Q_2 = 0$ and that X changes midway between the rising and falling clock edges.

c. List the output values produced by the input sequence.



$$Q_1^+ = J_1 Q_1' + K_1 Q_1$$

$$Z = XQ_1 + XQ_2$$

$$Q_2^+ = J_2 Q_1' + K_2 Q_2$$

$$Q_1^+ = XQ_1' + XQ_2 + XQ_1$$

$$Q_2^+ = XQ_1 + XQ_2 + XQ_2$$

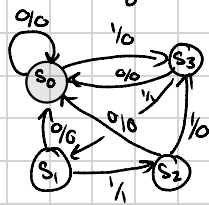
$$Q_1^+ = X + (XQ_2)'$$

$$Q_2^+ = (Q_1X)' + X$$

a) truth table

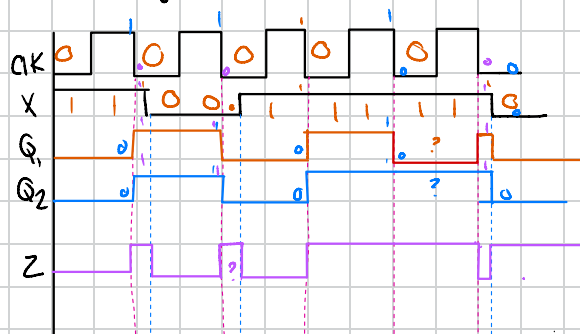
Q ₁ Q ₂	Q ₁ ⁺ Q ₂ ⁺		Z=0	Z=1
	X=0	X=1		
00	00	11	1	0
01	00	10	0	1
11	00	01	1	1
10	00	11	1	0

b) State graph



Note: for falling edge look at the time state before the line

c) time diagram

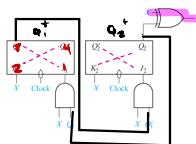


CLK	X	Q ₁
0	1	0
1	0	0
2	1	0
3	0	0
4	1	0
5	0	0
6	1	0
7	0	0
8	1	0
9	0	0
10	1	0
11	0	0
12	1	0
13	0	0
14	1	0
15	0	0

13.11 a. Construct a transition table and state graph for the given circuit.

b. Construct a timing chart for the circuit for an input sequence $X = 10011$.

Indicate at what times Z has the correct value and specify the correct output sequence. Assume that X changes midway between falling and rising clock edges.) Initially, $Q_1 = Q_2 = 0$.



Details

a)

Note: this is a mealy machine, so Z depends on X

$$Q_1^+ = J_1 Q_1' + K_1 Q_1$$

Self Note: inside of the mealy machine the variables are doing cross prime multiplying (13 + 2'4)

$$Q_1^+ = Q_1'XQ_2 + XQ_1$$

$$Q_2^+ = J_2 Q_1' + K_2 Q_2$$

$$Q_2^+ = Q_2'XQ_1 + XQ_2$$

$$Z = XQ_2 + Q_2'X$$

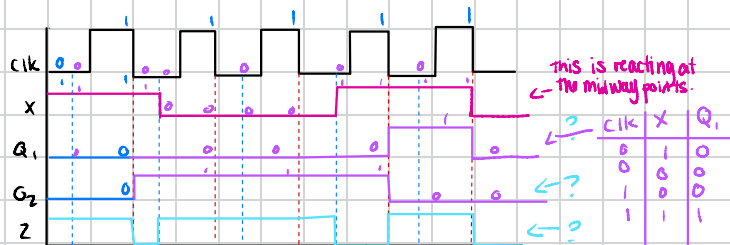
$$X=0=Q_1^+ \quad X=1=Q_2^+$$

Q ₁ Q ₂	0	1	Z=0	Z=1
00	00		0	1
01	01		1	0
11	11		1	0
10	10		0	1

X=0 for this machine keeps the same state.

X=10011

c) time diagram



this is reacting at the midway points.

CLK	X	Q ₁
0	1	0
1	0	0
2	0	0
3	1	0
4	0	0
5	0	0
6	1	0
7	0	0
8	0	0
9	1	0
10	0	0
11	0	0
12	1	0
13	0	0
14	0	0
15	1	0
16	0	0